CareBridge: A Local-First, Multilingual Medicine Companion for Patients After Hospital Discharge, with Row-by-Row Review and an Honest "Not Checked" Safety Layer

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***Abstract*—*Patients often go home with a discharge summary written in dense, abbreviated English, and medicine errors after discharge are a known cause of preventable harm. We present CareBridge, a prototype web application that reads a photographed discharge summary or prescription using a vision-language model run entirely on the user's own machine (Qwen3-VL 8B served through Ollama). Deterministic rules then assign each medicine a review level, a formulary picker lets the user confirm uncertain rows, and each medicine is explained in English, Tamil or Hindi with line-by-line verification. Medicine pairs are checked against DDInter, and a daily schedule is produced. The safety layer never declares a combination safe: any pair that cannot be checked is labelled* not checked*. We evaluate the pipeline under a frozen, pre-registered protocol on 40 held-out synthetic documents in five image conditions (795 medicine mentions). After the rule layer, medicine F1 improved from 0.984 to 0.999 and fully correct entries from 26.2% to 88.8%. No wrong generic name reached the interaction check (0 of 389), yet 51.1% of medicines needed user confirmation and 2 of 285 schedule-ready medicines received a wrong schedule. English explanation lines passed verification 93.7% of the time, Hindi 66.7% and Tamil only 18.0%. A stress set of 30 authored prescriptions satisfied all four pre-registered criteria. We report failures, document-level confidence intervals and limitations in full: the data are synthetic, nothing is clinically validated, and no native speaker has reviewed the Tamil or Hindi text.*

***Index Terms*—*post-discharge care, medication safety, vision-language model, multilingual health communication, drug–drug interaction, local-first AI*

I. Introduction

A hospital stay usually ends with a sheet of paper that names each drug by brand or generic name and relies on shorthand such as OD (once daily), BD (twice daily), PC (after food) and SOS (only if needed). Adverse events are common after discharge, and many are linked to medicines [1]; the World Health Organization has chosen medication safety as the theme of its third Global Patient Safety Challenge [2]. Limited health literacy is tied to poorer outcomes [3], and instructions in a second language add a further obstacle for patients and caregivers in multilingual regions such as India.

Vision-language models (VLMs) can now read a photographed page directly [5], which points toward a tool that patients could use themselves. The obstacle is trust. A VLM may quietly "repair" a misspelt drug name, copy text into an empty field, or write fluent explanations that nothing supports [7]. A tool that is wrong in a confident tone may do more harm than no tool.

This paper describes CareBridge, a class-project prototype designed around a single rule: when the system is unsure it must say so, and it must never claim that a combination is safe. Our contributions are as follows.

1. A local-first pipeline from photograph to explanation, interaction check and schedule, in which each medicine carries a review level (`must_confirm`, `note` or `none`).
2. A safety engine with four statuses, in which every pair that cannot be checked is reported as *not checked* together with the reason.
3. A frozen, single-run evaluation on held-out synthetic documents, reporting raw and after-rules results, intervals that respect document clustering, a post-hoc case review, and a 30-prescription stress test.
4. A candid account of what went wrong, including a weak Tamil pipeline and two incorrect schedules.

II. Related Work

Safety after discharge and health literacy. Forster et al. showed that patients continue to be harmed after leaving hospital, with drug-related events among the most frequent [1]. Berkman et al. connected low health literacy with worse outcomes [3]. These studies establish the need but offer no tool.

Reading and generating text. Open VLMs such as Qwen3-VL interpret document images without a separate OCR stage [5] and can be hosted locally [6]. Large language models hold substantial clinical knowledge [9] but are prone to hallucination [7]. LLM judges agree with human raters on average, yet show biases, including a preference for text produced by the same model [8]; our pipeline uses model-based verification and we acknowledge this bias.

Interaction knowledge. DDInter is an open, curated database of drug–drug interactions with severity grades [4]. Its coverage

is incomplete, so an interface must keep "interaction found", "no record" and "not confirmed" apart.

Statistics for small samples. The Wilson interval [10] and the bootstrap [11] suit small samples. Because medicines from one document are not independent, we also resample whole documents.

III. System Design

A. Architecture and Pipeline

CareBridge consists of a React front end and a FastAPI back end. Documents and results are stored in SQLite with Fernet-encrypted payloads, and the model runs under Ollama on the same machine (Fig. 1), so the real workflow needs no cloud service. A public GitHub Pages site displays only a pre-recorded sample, with no server and no `/api` calls.

The pipeline has six stages. (1) *Extraction*: the model receives the image (longest side 1,280 px) and a fixed prompt, and returns one JSON object per medicine with a verbatim quote for every field (temperature 0, seed 42). (2) *Normalisation*: rules separate fields that mix brand, generic and form, repair role confusions, match names against a 66-drug formulary, and compare each value with its quote. (3) *Confirmation*: medicines at `must_confirm` are held back until the user selects the intended drug from the formulary. (4) *Explanation*: plain-language lines of three kinds (advice, follow-up, warning) are drawn from templates and a phrase library, with a verified model-generated path. (5) *Safety check*. (6) *Scheduling*: medicines marked `schedule_ready` are placed on a daily plan, and a one-page doctor summary is produced.

B. Review Levels and Readiness

Every medicine receives one review level and two readiness flags. The flag `interaction_ready` means the name has been matched to a formulary generic and confirmed; `schedule_ready` means frequency and timing are unambiguous. The flags `unresolved_name`, `duplicate_name_fields`, `role_corrected`, `incomplete_quote` and `unusual_strength` keep a medicine out of interaction checks. The flag `ambiguous_OD` (OD written on an eye drop can mean once daily or right eye) and the weekly-medicine flag block a daily schedule, while `conflict_timing_label` appears only as a note.

C. Interaction-Safety Engine

For each pair of medicines the engine returns exactly one of four statuses (Fig. 2): `not_checked_unconfirmed`, `not_checked_database_missing`, `not_checked_no_record`, or `interaction_found` (with level, source and advice). Among the 2,145 pairs formed by the 66 formulary drugs, 1,014 have a record (998 from DDInter and 16 from an author-checked supplement) and 1,131 have none. Of those records, 17 are major, 261 moderate, 44 minor and 692 of unknown severity. The word "safe" occurs only in negated form.

D. Privacy

Sessions are anonymous, a versioned consent notice appears before any upload, state-changing calls need a custom request header, user content is never cached on disk, and a "delete my data" control empties every table (in about 0.4 s, even while a background job is running).

IV. Evaluation Protocol

A. Data

Real discharge papers are private and have no ground truth, so every document here is synthetic, generated from templates with known medicines. The development split holds 20 documents. The held-out test split holds 40 documents, each rendered as a clean image, plain text, a scan, a phone photo and a poor-quality photo, giving 200 readings and 795 medicine mentions. The test split was left unopened while the harness was developed. A stress set of 30 authored prescriptions spans ten categories (major, moderate and supplement pairs, duplicates, garbled names, weekly medicines, unusual strengths, eye drops with OD, an eight-medicine list, and controls), three of each, in clean and photo renders. Development and test data come from a single generator, so the test split is held out from tuning but is not independent of the generator's style. It also contains more clean documents (16 clean, 11 abbreviated, 13 messy) than the development split (5, 9, 6).

B. Freeze and Single Run

Before testing, the configuration was frozen using a git tag, a manifest hashing 98 files, the model digest (Qwen3-VL 8B instruct) and a hash of the test data. The runner refuses an official run if the working tree is dirty, if anything has drifted, or if an official run already exists. The settings were Ollama 0.40.5, context 6,144, at most 3,000 new tokens, temperature 0 and seed 42. The run took roughly 2 h 25 min on a laptop with an 8 GB GPU.

C. Metrics and Statistics

We report medicine precision, recall and F1; the fully correct rate (all nine fields right); hallucinated medicines; name errors that remain; the share at `must_confirm`; the number of ready medicines; wrong generic names among interaction-ready medicines (target 0); wrong schedules; strict silent misses (a true interaction neither shown nor covered by a flag); false alarms; and explanation verification by language. Each is given raw and after rules. Proportions use Wilson 95% intervals, while F1 and the fully correct rate use a seeded document-level bootstrap (2,000 resamples). A post-hoc review after the run re-examined failures without re-running the model and is labelled as such.

V. Results

A. Extraction

Table 1 gives pooled results across the five conditions. The raw model finds almost every medicine (F1 0.984), yet only about a quarter of entries are fully correct. The main reason is that it fills empty brand and generic fields with text from elsewhere on the page (276 of 313 empty brand fields were filled, 269 of them with another name from the same document), and it sometimes places several names in one field. The rule layer raises fully correct entries to 88.8% and timing-label accuracy from 0.734 to 0.990. Remaining name errors increase as image quality falls: 1 (text), 2 (clean), 4 (scan), 7 (photo) and 12 (poor), 26 of 795 in total (Fig. 3).

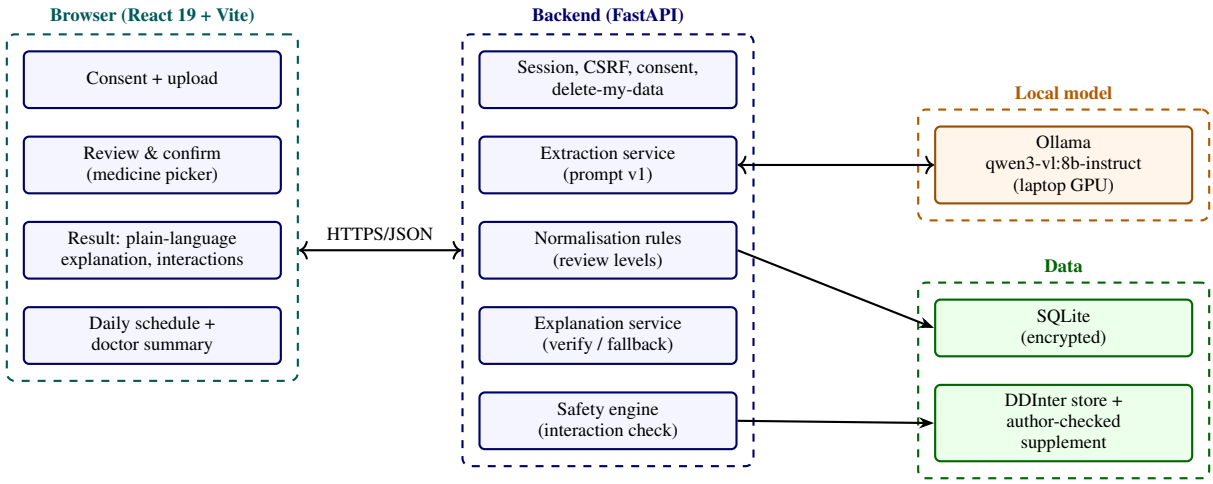


Figure 1. System architecture. Every component of the real workflow runs on the user's own laptop.

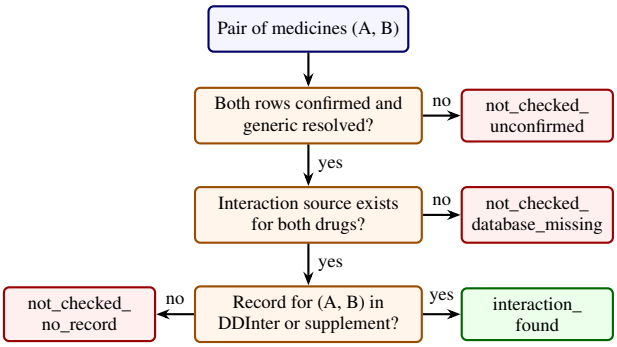


Figure 2. Decision flow of the safety engine. No branch ends in "safe"; "not checked" means no information, not no risk.

Table 1
POOLED RESULTS ON THE HELD-OUT TEST (95% INTERVALS)

Measure	Raw output	After rules
Medicine F1	0.984 [0.972, 0.993]	0.999 [0.996, 1.000]
Fully correct	0.262 [0.200, 0.328]	0.888 [0.845, 0.927]
Hallucinated medicines	13	1
Share at `must_confirm`	0.511 [0.476, 0.545]	
`interaction_ready`	389 of 795	
`schedule_ready`	285 of 795	
Wrong generic (interaction-ready)	0 of 389	
Wrong schedule (schedule-ready)	2 of 285	

B. Safety

No wrong generic name reached the interaction check (0 of 389 readings; rule-of-three upper bound 0.8% per reading), and there were no false alarms. Three strict silent misses occurred, but all stem from one misread row in one document ("Performin" read for Metformin). That row was held at `must_confirm`, so its pairs were reported as not checked instead of being hidden, and none escaped a flag. Between 79% and 89% of pairs were *not checked*, chiefly because more than half of formulary pairs have no record in either source.

Two schedules were wrong, both in scanned documents. In one, SOS for an inhaler was read as OD and the quote agreed with the misreading, so no flag fired. In the other, the token PC was dropped from "BD PC", leaving only a note-level conflict that did not block scheduling. These are the main residual risks: a misreading consistent with its own quote cannot be caught by rules, and a note-level conflict does not stop a schedule.

Table 2
EXPLANATION LINES VERIFIED, BY LANGUAGE AND LINE KIND

Language	Verified	Fallback	Advice	Follow-up	Warning
English	0.937	0.004	0.873	0.976	1.000
Hindi	0.667	0.275	0.754	0.762	0.516
Tamil	0.180	0.761	0.254	0.286	0.042

C. Explanations in Three Languages

Table 2 and Fig. 4 show verification over 255 lines per language. English performs well, Hindi is partial and weakest on warnings, and Tamil is poor: 194 of 255 lines failed verification and were displayed in English with a notice, mostly because the judge found the text contradicted or omitted facts. Two of 320 explanation calls were truncated (both Tamil). Because the same model writes and judges, and no native speaker has reviewed any text, these figures describe this pipeline and not translation quality in general.

D. Development versus Test, and Interval Width

Test results were worse than development on raw brand-name accuracy (0.673 down to 0.495) and on the `must_confirm` share (0.462 up to 0.511), and better on after-rules fully correct entries (0.811 up to 0.888). The test set is easier in its document mix: re-weighting to a common difficulty mix narrows the after-rules gap from $+0.076$ to about $+0.044$ but does not remove it, so the comparison cannot rule out overfitting. Wilson intervals over pooled mentions are too tight because medicines cluster within documents; a document-level bootstrap roughly doubles their width (for example, `must_confirm` 0.511 has Wilson [0.476, 0.545] but cluster [0.436, 0.578]; after-rules fully correct 0.888 has [0.864, 0.908] against [0.846, 0.926]). For wrong generic names, the exact one-sided 95% bound across 39 documents is 7.4%.

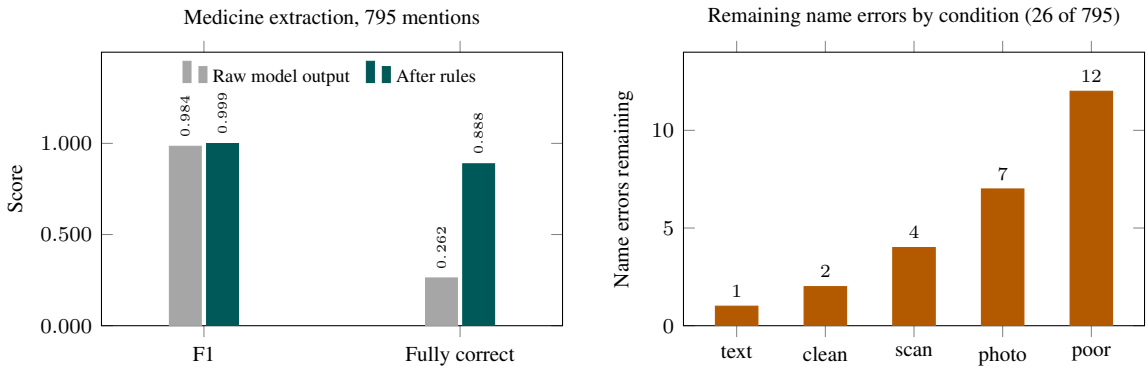


Figure 3. Left: raw versus after-rules extraction (95% intervals in Table 1). Right: name errors left after the rules, by image condition.

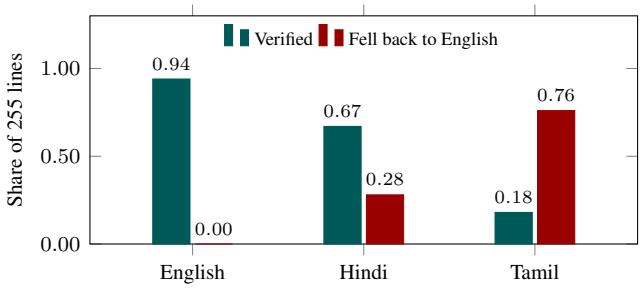


Figure 4. Share of explanation lines that were verified versus replaced by English, by language.

E. Stress Test

The stress extraction (60 calls, 29 min, run on Ollama 0.40.11, a recorded departure from the pinned 0.40.5) met all four pre-registered criteria: 0 of 204 silent misses lacked a flag, 0 of 6 weekly medicines landed on a daily schedule, 0 of 113 interaction-ready medicines had a wrong generic, and 204 of 204 pairs were correct after scripted confirmation. All 104 medicines were found in both renders with no hallucinations, and every duplicate, weekly medicine and eye drop with OD was flagged. The price of caution is visible: only 38 of 102 ground-truth pairs per render (37.3%, Wilson [0.285, 0.469]) were shown at once, because flags on correctly read medicines held back the other 64 until confirmation; with perfect extraction the engine would show 100 of 102. One unflagged silent correction ("Ibuirofen" to "Ibuprofen", also in the quote) was held back only by an unrelated flag.

VI. Discussion and Limitations

Layering helps: the model alone had high F1 but a low fully correct rate, while the rules plus confirmation kept wrong drug identities out of the safety check. The cost is a heavy review burden (about half of all medicines) and mostly "not checked" pairs, so the tool is honest but not yet convenient. Tamil support is weak, and some errors are invisible to rules.

Limitations. (i) All data are synthetic and come from one generator; real-photo and usability studies were prepared but not run. (ii) The system is not clinically validated and is not a medical device; the interaction supplement and all Tamil and Hindi strings have had no pharmacist or native-speaker review. (iii) The same model writes and judges explanations [8]. (iv) Model output is not reproducible across Ollama versions (flags differed in a check on five documents). (v) With 40 test documents, intervals are wide. (vi) Misreadings that agree with their quote, and note-level timing conflicts, can still reach a schedule.

Future work. Real-photo and usability studies; pharmacist and native-speaker review; an independent check of form against frequency, or a second reading; blocking schedules on timing conflicts; broader interaction sources; difficulty-stratified splits and an independent test generator.

VII. Conclusion

CareBridge demonstrates that a local, multilingual medicine companion can be built around uncertainty. A rule layer with row-level confirmation lifted fully correct medicine entries from 26.2% to 88.8% and kept wrong drug identities out of the interaction check, while every pair that could not be verified remained *not checked*. The evaluation also exposes real weaknesses: two wrong schedules, a high confirmation burden, and Tamil text that mostly falls back to English. We release the code, protocol and frozen results (MIT licence; `https://github.com/ItsTheDemiGod/carebridge`) together with a pre-recorded demonstration (`https://itsthedemigod.github.io/carebridge/`). CareBridge is a prototype and must not be used to make treatment decisions.

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